Base ESP32-S3 wiring cheat sheet

Battery unplugged before any wire moves. ESP32-S3 DevKitC-1-N8, powered only by USB (port marked USB) from the Jetson or the Mac. Pins from `firmware/base/WIRING.md` (that file wins if this page is older).

GPIO	Name in pins.h	Connects to	Note
4	`PIN_RPWM`	BTS7960 RPWM	wiper FORWARD (dir 1), PWM 20 kHz
5	`PIN_LPWM`	BTS7960 LPWM	wiper BACKWARD (dir -1), PWM 20 kHz
6	`PIN_REN`	BTS7960 R_EN	plus 10 k from R_EN to GND
7	`PIN_LEN`	BTS7960 L_EN	plus 10 k from L_EN to GND
1	`PIN_WIPER_IS`	BTS7960 R_IS + L_IS through 1 k	plus 680 R from GPIO1 to GND (the IS divider)
2	`PIN_VBAT`	battery divider middle	100 k from 3S + (at the BTS7960 B+ screw) to GPIO2, 22 k from GPIO2 to GND
8	`PIN_SDA`	AS5600 SDA	STEMMA QT blue wire
9	`PIN_SCL`	AS5600 SCL	STEMMA QT yellow wire
15	`PIN_DRUM_IN1`	DRV8833 AIN1	drum N20
16	`PIN_DRUM_IN2`	DRV8833 AIN2	drum N20
10	`PIN_SW_A`	contact `SW_DRUM_A`	drum code bit A
11	`PIN_SW_B`	contact `SW_DRUM_B`	drum code bit B
12	`PIN_SW_C`	contact `SW_DRUM_C`	drum code bit C
13	`PIN_SW_D`	contact `SW_DOG_L`	collar engaged in LAUNCH
14	`PIN_SW_E`	contact `SW_DOG_R`	collar engaged in REEL
47	`PIN_SW_F`	contact `SW_N20`	drum N20 in mesh (open = hand lever latched out)
21	`PIN_SW_G`	contact `SW_HOME`	carriage on its front stop (uncocked)
39	`PIN_SW_H`	contact `SW_GAP`	big gear in its toothless gap (rest)
40	`PIN_SW_I`	contact `SW_RAIL_LOW`	rail at its lowest pitch (aim home)
41	`PIN_SW_J`	contact `SW_SEATED`	dart tail on the pusher
42	`PIN_SW_K`	contact `SW_ARM`	wrap cup ring at home (one pulse per cup turn)
35	`PIN_SW_L`	contact `SW_INDEX`	only on a build without the AS5600 magnet (decision D12). Leave open otherwise
17	`PIN_HX_DT`	HX711 DT (DOUT)	optional load cell
18	`PIN_HX_SCK`	HX711 SCK	optional load cell
3V3		AS5600 VIN, BTS7960 VCC, HX711 VCC	logic supply
GND		common ground (see section 2)	

Power: 3S LiPo, XT60 lead #2 (20 A fuse), BTS7960 B+ / B- (14 AWG). Wiper on M+ (red) / M- (black), 14 AWG. MP1584 #2 from the B+ / B- screws through a 2 A inline fuse, set to 9.0 V BEFORE its output is connected, out to DRV8833 VM / GND. DRV8833 #3: only AIN1, AIN2, GND go to the ESP32; SLP is not wired (decision D9).

Resistors: 10 k R_EN to GND, 10 k L_EN to GND, IS: R_IS + L_IS tied, 1 k to GPIO1, 680 R GPIO1 to GND. Battery: 100 k at the B+ screw to GPIO2, 22 k GPIO2 to GND. Common ground: ESP32 GND, BTS7960 GND, DRV8833 GND, MP1584 OUT-.

Never: the 3S on the DRV8833 VM (10.8 V max); 3V3 on a pad that beeps to VM; a plug pulled with the battery in.